

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030597

NAME: U.VIDHATH

Query 1: Create the database

```
CREATE DATABASE IF NOT EXISTS sql_db;  
USE sql_db;
```

Output: -

✓	3	09:44:54	CREATE DATABASE IF NOT EXISTS sql_db	1 row(s) affected	0.000 sec
✓	4	09:44:58	USE sql_db	0 row(s) affected	0.016 sec

Query 2: -CREATE PHYSICIAN TABLE

```
-- 2520030001-Aswitha  
CREATE TABLE 2520030597-Vidhath  
    employeeid INT PRIMARY KEY,  
    name VARCHAR(100),  
    position VARCHAR(100),  
    ssn VARCHAR(20)  
);
```

Output: -

✓	5	09:46:21	CREATE TABLE physician (employeeid INT PRI...	0 row(s) affected	0.063 sec
---	---	----------	--	-------------------	-----------

Query 3: -INSERT DATA INTO PHYSICIAN

2520030597-Vidhath

INSERT INTO physician VALUES

```
(1,'M.Vidhath','Staff Internist','11111111'),
(2,'M.Raghuveer','Attending Physician','22222222'),
(3,'A.Durga prasad','Surgical Attending Physician','33333333'),
(4,'A.Vishnu','Senior Attending Physician','44444444'),
(5,'Adithya Vardhan','Head Chief of Medicine','55555555'),
(6,'Ananya varanasi','Surgical Attending Physician','66666666'),
(7,'Yashwanth Reddy','Surgical Attending Physician','77777777'),
(8,'Nihitha','MD Resident','88888888'),
(9,'Ramya Sree','Attending Psychiatrist','99999999');
```

Output: -

```
9 14:32:03 INSERT INTO physician VALUES (1,'M.Aswitha P... 9 row(s) affected Records: 9 Duplicates: 0 Wamin... 0.016 sec
```

Query 4: -CREATE DEPARTMENT TABLE

```
CREATE TABLE department (
    departmentid INT PRIMARY KEY,
    name VARCHAR(100),
    head INT
);
```

Output: -

```
10 14:33:35 CREATE TABLE department ( departmentid INT ... 0 row(s) affected 0.047 sec
```

Query 5: -INSERT DATA INTO DEPARTMENT

2520030597-Vidhath

```
(1,'General Medicine',7),
(2,'Surgery',7),
(3,'Psychiatry',9);
```

Output: -

```
11 14:35:14 INSERT INTO department VALUES (1,'General Me... 3 row(s) affected Records: 3 Duplicates: 0 Wamin... 0.016 sec
```

Query 6: -INNER JOIN

2520030597-Vidhath

```
SELECT d.name AS Department,  
       p.name AS Head_Physician  
FROM department d  
JOIN physician p  
ON d.head = p.employeeid;
```

Output: -

✓	12	14:36:41	SELECT d.name AS Department, p.name AS ...	3 row(s) returned	0.000 sec / 0.000 sec
Result Grid					
	Department	Head_Physician			
▶	General Medicine	Vishnu			
	Surgery	Yashwanth Reddy			
	Psychiatry	Ramya Sree			

Query 7: - WHERE CLAUSE

2520030597-Vidhath

```
SELECT *  
FROM physician  
WHERE position='Surgical Attending Physician';
```

Output: -

✓	13	14:38:23	SELECT * FROM physician WHERE position='Sur...	3 row(s) returned	0.016 sec / 0.000 sec
Result Grid					
	employeeid	name	position	ssn	
▶	3	Durga prasad	Surgical Attending Physician	333333333	
	6	Varshith Reddy	Surgical Attending Physician	666666666	
	7	Yashwanth Reddy	Surgical Attending Physician	777777777	
*	NULL	NULL	NULL	NULL	

Query 8: - WHERE + LIKE

2520030597-Vidhath

```
SELECT *  
FROM physician  
WHERE name LIKE 'Nihitha%';
```

Output: -

15 14:40:11 SELECT * FROM physician WHERE name LIKE '...' 1 row(s) returned 0.000 sec / 0.000 sec

	employeeid	name	position	ssn
▶	8	Nihitha	MD Resident	888888888
*	NULL	NULL	NULL	NULL

Query 9: - COUNT()

2520030597-Vidhath

```
SELECT COUNT(*) AS Total_Physicians  
FROM physician;
```

Output: -

16 14:42:16 SELECT COUNT(*) AS Total_Physicians FROM p... 1 row(s) returned 0.032 sec / 0.000 sec

	Total_Physicians
▶	9

Query 10: - COUNT DISTINCT

2520030597-Vidhath

```
SELECT COUNT(DISTINCT position)  
FROM physician;
```

Output: -

17 14:44:11 SELECT COUNT(DISTINCT position) FROM phys... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	Filter Rows:
COUNT(DISTINCT position)	
7	

Query 11: -GROUP BY

```
SELECT position,
        COUNT(*) AS Total_Employees
FROM physician
GROUP BY position;
```

Output: -

18
14:47:37
SELECT position, COUNT(*) AS Total_Emplo...
7 row(s) returned
0.016 sec / 0.000 sec

	position	Total_Employees
▶	Staff Internist	1
	Attending Physician	1
	Surgical Attending Physician	3
	Senior Attending Physician	1
	Head Chief of Medicine	1
	MD Resident	1
	Attending Psychiatrist	1

Query 12: -GROUP BY + HAVING

```
SELECT position,
        COUNT(*) AS Total_Employees
FROM physician
GROUP BY position
HAVING COUNT(*) > 1;
```

Output: -

19	14:49:06	SELECT position, COUNT(*) AS Total_Employees	1 row(s) returned	0.000 sec / 0.000 sec
----	----------	--	-------------------	-----------------------

Result Grid			Filter Rows:	Export:
	position	Total_Employees		
▶	Surgical Attending Physician	3		

Query 13: -ORDER BY

```
SELECT *
FROM physician
ORDER BY name ASC;
```

Output: -

✓	20	14:50:34	SELECT * FROM physician ORDER BY name AS...	9 row(s) returned	0.032 sec / 0.000 sec
---	----	----------	---	-------------------	-----------------------

Result Grid					Filter Rows:	Edit:	Export:
	employeeid	name	position	ssn			
▶	5	Adithyan Vardhana	Head Chief of Medicine	555555555			
	3	Durga prasad	Surgical Attending Physician	333333333			
	6	Varshith Reddy	Surgical Attending Physician	666666666			
	4	Vishnu	Senior Attending Physician	444444444			
	2	M.Raghuveer	Attending Physician	222222222			
	1	M.Vidhath	Staff Internist	111111111			
	8	Nihitha	MD Resident	888888888			
	9	Ramya Sree	Attending Psychiatrist	999999999			
	7	Yashwanth Reddy	Surgical Attending Physician	777777777			
*	NULL	NULL	NULL	NULL			

Query 14: -ORDER BY DESC

2520030597-Vidhath

```
SELECT *
FROM physician
ORDER BY employeeid DESC;
```

Output: -

✓	20	14:50:34	SELECT * FROM physician ORDER BY name AS...	9 row(s) returned	0.032 sec / 0.000 sec
---	----	----------	---	-------------------	-----------------------

	employeeid	name	position	ssn
▶	5	Adithya Vardhan	Head Chief of Medicine	555555555
	3	Durga prasad	Surgical Attending Physician	333333333
	6	Varshith Reddy	Surgical Attending Physician	666666666
	4	Vishnu	Senior Attending Physician	444444444
	2	M.Raghuveer	Attending Physician	222222222
	1	M.Vidhath	Staff Internist	111111111
	8	Nihitha	MD Resident	888888888
	9	Ramya Sree	Attending Psychiatrist	999999999
	7	Yashwanth Reddy	Surgical Attending Physician	777777777
*	NULL	NULL	NULL	NULL

Query 15: - WHERE + IN

```

SELECT *
FROM physician
WHERE employeeid IN
(
    SELECT head
    FROM department
);

```

Output: -

21 14:54:58 SELECT * FROM physician WHERE employeeid I... 3 row(s) returned 0.000 sec / 0.000 sec

Result Grid				
	employeeid	name	position	ssn
▶	4	Anirudh Reddy	Senior Attending Physician	444444444
	7	Yashwanth Reddy	Surgical Attending Physician	777777777
	9	Ramya Sree	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

Query 16: -SUBQUERY

2520030597-Vidhath

```
SELECT *  
FROM physician  
WHERE employeeid NOT IN  
(  
    SELECT head  
    FROM department  
);
```

Output: -

✓ 22 14:56:50 SELECT * FROM physician WHERE employeeid ... 6 row(s) returned 0.032 sec / 0.000 sec

	employeeid	name	position	ssn
▶	1	M.Vidhath	Staff Internist	111111111
	2	M.Raghuveer	Attending Physician	222222222
	3	Durga prasad	Surgical Attending Physician	333333333
	5	Adithya Vardhan	Head Chief of Medicine	555555555
	6	Varshith Reddy	Surgical Attending Physician	666666666
	8	Nihitha	MD Resident	888888888
★	NULL	NULL	NULL	NULL

Query 17: - GROUP BY + ORDER BY

2520030597-Vidhath

```
SELECT position,  
       COUNT(*) AS Total_Employees  
FROM physician  
GROUP BY position  
ORDER BY Total_Employees DESC;
```

Output: -

✓ 23 14:58:30 SELECT position, COUNT(*) AS Total_Emplo... 7 row(s) returned 0.000 sec / 0.000 sec

2520030597-Vidhath		employees
☒	Surgical Attending Physician	3
☐	Staff Internist	1
☐	Attending Physician	1
☐	Senior Attending Physician	1
☐	Head Chief of Medicine	1
☐	MD Resident	1
☐	Attending Psychiatrist	1

Query 18: - JOIN + WHERE

```
2520030597-Vidhath
SELECT p.name
FROM physician p
JOIN department d
ON p.employeeid=d.head
WHERE d.name='Surgery';
```

Output: -

✓	24	15:01:13	SELECT p.name FROM physician p JOIN depart...	1 row(s) returned	0.000 sec / 0.000 sec
---	----	----------	---	-------------------	-----------------------

name
Yashwanth Reddy

Query 19: - JOIN + ORDER BY

```
-- 2520030001-Aswitha
SELECT d.name,
       p.name
FROM department d
JOIN physician p
ON d.head=p.employeeid
ORDER BY d.name;
```

Output: -

✓	25	15:02:46	SELECT d.name, p.name FROM department ...	3 row(s) returned	0.000 sec / 0.000 sec
---	----	----------	---	-------------------	-----------------------

	name	name
▶	General Medicine	Vishnu
	Psychiatry	Ramya Sree
	Surgery	Yashwanth Reddy

Query 20: - WHERE + NOT LIKE

```
2520030597-Vidhath
FROM physician
WHERE position NOT LIKE '%Surgical%';
```

Output: -

✓ 26 15:04:17 SELECT * FROM physician WHERE position NO... 6 row(s) returned 0.000 sec / 0.000 sec

	employeeid	name	position	ssn
▶	1	M.Vidhath	Staff Internist	111111111
	2	M.Raghuveer	Attending Physician	222222222
	4	Vishnu	Senior Attending Physician	444444444
	5	Adithya Vardhan	Head Chief of Medicine	555555555
	8	Nihitha	MD Resident	888888888
	9	Ramya Sree	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

Query 21: - COUNT + HAVING

```
2520030597-Vidhath
SELECT position,
       COUNT(*) AS Total
FROM physician
GROUP BY position
HAVING COUNT(*) >= 2;
```

Output: -

✓ 27 15:05:45 SELECT position, COUNT(*) AS Total FROM ... 1 row(s) returned 0.000 sec / 0.000 sec

	position	Total
▶	Surgical Attending Physician	3

Query 22: - EXISTS

2520030597-Vidhath

```
FROM department d
WHERE EXISTS
(
    SELECT 1
    FROM physician p
    WHERE p.employeeid=d.head
);
```

Output: -

✓ 28 15:07:03 SELECT * FROM department d WHERE EXISTS ... 3 row(s) returned 0.000 sec / 0.000 sec

2520030597-Vidhath

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
*	NULL	NULL	NULL

Query 23: - MAX()

```
-- 2520030001-Aswitha
SELECT MAX(employeeid)
FROM physician;
```

Output: -

✓ 29 15:08:41 SELECT MAX(employeeid) FROM physician LIMIT... 1 row(s) returned 0.000 sec / 0.000 sec

	MAX(employeeid)
▶	9

Query 24: - MIN()

```
-- 2520030001-Aswitha
SELECT MIN(employeeid)
FROM physician;
```

Output: -

✓ 30 15:10:24 SELECT MIN(employeeid) FROM physician LIMIT... 1 row(s) returned 0.000 sec / 0.000 sec

Result Grid	
	MIN(employeeid)
▶	1